Review of
Scientific Instruments

Molecular Beam Reactive Scattering Apparatus with Electron Bombardment Detector

Y. T. Lee, J. D. McDonald, P. R. LeBreton, and D. R. Herschbach

Citation: Review of Scientific Instruments **40**, 1402 (1969); doi: 10.1063/1.1683809
View online: http://dx.doi.org/10.1063/1.1683809
View Table of Contents: http://scitation.aip.org/content/aip/journal/rsi/40/11?ver=pdfcov
Published by the AIP Publishing

Thus, the voltage recorded at the oscilloscope is the voltage produced by the gauge but delayed by the transit time of the cable.

In other words, the voltage at the gauge initially sees a voltage divider consisting of the input resistance and cable impedance. It therefore transits the cable at one-half the gauge voltage. Upon arriving at the oscilloscope the pulse sees an open circuit and thus doubles its voltage propagating back along the cable as a reflected wave. The reflected wave upon reaching the gauge is properly terminated. In practice there is a problem in that R does not maintain a constant value during the transit of the stress profile. It has been found that if the resistance is chosen such that proper termination is effected when the gauge reaches its final state value this problem is eliminated.

ACKNOWLEDGMENTS

The author wishes to thank Dr. G. R. Fowles under whose supervision this work was performed. He also would like to thank Tom Michaels for valuable advice on electronics.

THE REVIEW OF SCIENTIFIC INSTRUMENTS VOLUME 40, NUMBER 11 NOVEMBER 1969

Molecular Beam Reactive Scattering Apparatus with Electron Bombardment Detector

Y. T. Lee,* J. D. McDonald,† P. R. LeBreton, and D. R. Herschbach‡

Department of Chemistry, Harvard University, Cambridge, Massachusetts 02138

(Received 20 June 1969; and in final form, 22 July 1969)

A crossed beams apparatus for study of thermal energy neutral–neutral reactions is described. The detector comprises a high efficiency (~0.1%) electron bombardment ionizer, quadrupole mass filter, scintillation ion counter, and gated scalers synchronized with the beam modulation. The ionizer is nested within three chambers, each pumped by a separate ion pump and the innermost attached to a liquid nitrogen trap. The design is such that molecules which pass through the ionizer without being ionized fly on into another differentially pumped region before hitting a surface. The entire detector unit, including pumps and trap, is mounted on a rotatable platform (angular range ~140°) which forms the lid of the scattering chamber. The beam sources also comprise modular units, mounted in differentially pumped side chambers which insert into ports in the scattering chamber. Angular distributions of reaction products have been measured for several reactions of Cl, Br, or D atoms with halogen molecules or hydrogen halides. Product velocity distributions have also been measured by time of flight for some of these reactions. In most cases, the partial pressure of interfering background species in the ionization region was $\lesssim 10^{-15}$ Torr, and satisfactory data could be obtained with reactive scattering signals of a few counts per second.

SCATTERING experiments with crossed molecular beams offer the most direct means to study the dynamics of chemical reactions. For more than a decade such studies have been pursued in several laboratories. This work has given much detailed information about angular distributions of products; disposal of reaction exothermicity among translation, rotation, and vibration; lifetime of the intermediate collision complex; variation of reaction cross section with collision energy, impact parameter, and orientation of the target molecule; and the correlation of reaction dynamics with electronic structure.[1] However, because of the difficulty of detecting reaction products under molecular beam conditions, until recently all successful neutral–neutral reaction studies had been limited to systems involving alkali species, for which surface ionization provides a uniquely sensitive and specific detector. The low ionization potential of alkali atoms, which makes surface ionization feasible, also makes ionic bonding dominant in these reactions whereas covalent bonding governs most chemical reactions.

In molecular beam spectroscopy a "universal" detector employing electron bombardment ionization and mass analysis has been widely used.[2–7] Total cross sections and small angle elastic scattering for hydrogen and rare gas systems have been measured with such a detector.[8]

* Present address: James Franck Institute, University of Chicago, Chicago, Ill.

† National Science Foundation predoctoral fellow.

‡ During 1968–69, visiting fellow at the Joint Institute for Laboratory Astrophysics of the National Bureau of Standards and University of Colorado, Boulder, Colo.

1 For recent reviews, see (a) E. F. Greene and J. Ross, Science **159**, 587 (1968); (b) J. P. Toennies, Ber. Bunsenges. Physik. Chem. **72**, 927 (1968); for kindred ion-molecule work, see (c) B. H. Mahan, Acct. Chem. Res. **1**, 217 (1968); (d) R. Wolfgang, ibid. **2**, 248 (1969).

2 G. Wessel and H. Lew, Phys. Rev. **92**, 641 (1953).

3 W. E. Quinn, A. Pery, J. M. Baker, H. R. Lewis, N. F. Ramsey, and J. T. La Tourrette, Rev. Sci. Instrum. **29**, 935 (1958).

4 R. Weiss, Rev. Sci. Instrum. **32**, 397 (1961).

5 H. G. Bennewitz and R. Wedemeyer, Z. Physik **172**, 1 (1963).

6 G. O. Brink, Rev. Sci. Instrum. **37**, 857, 1626 (1966).

7 M. Kaufman, Ph.D. thesis, Harvard University, 1964; J. S. Muenter, M. Kaufman, and W. Klemperer, J. Chem. Phys. **48**, 3338 (1968).

8 (a) R. Duren, R. Feltgen, W. Gaide, R. Helbing, and H. Pauly, Phys. Lett. **18**, 282 (1965); **20**, 501 (1966); R. Duren, R. Helbing, and H. Pauly, Z. Physik **188**, 468 (1965); (b) G. E. Moore, S. Datz, and F. van der Walk, J. Chem. Phys. **46**, 2012 (1967); (c) J. Penta, C. R. Mueller, W. Williams, R. Olson, and P. Chakraborti, Phys. Lett. **25A**, 658 (1967); W. Williams, C. R. Mueller, P. McGuire, and B. Smith, Phys. Rev. Lett. **22**, 121 (1969).

Preliminary attempts to study reactive scattering of $D+H_2$, $H+D_2$, and $D+Br_2$ have also been reported.[9-11] Products were detected, but the interfering background was extremely high and the dynamical features of interest could not be determined despite use of very long counting times.

In the apparatus described in this paper, the background problem has been overcome by means of elaborate differential pumping and special design of the electron bombardment ionizer. Product angular distributions have been measured with this apparatus for the $Cl+Br_2$ and $D+Br_2$ reactions and for several others,[12,13] with results comparable in quality to those previously obtained for alkali systems. Studies of the $Cl+Br_2$ reaction in excellent agreement with our measurements have now been reported by two other laboratories.[14,15] The apparatus used by these groups evidently performs at least as well as ours, but both employ the fixed detector, movable ovens configuration rather than the movable detector, fixed ovens configuration we use, which is expected to be better suited to work with beams of hydrogen atoms and other species that require heavy pumping.

DESIGN CRITERIA

Apart from the small product yield and the inefficiency of the electron bombardment detector, the basic problem is suppression of the background which reaches the ionization region. In contrast to the alkali case, cryogenic trapping is not an adequate solution for species such as hydrogen or halogen atoms or molecules because of their high vapor pressure and the ease with which they desorb from surfaces. Even with liquid helium pumping, which was employed in the early work with hydrogen beams,[9-10] it is found that a layer of "hydrogen ice" quickly forms and incoming atoms and molecules are reflected from this surface with high probability. For such species the concentration of product molecules in the ambient gas within the scattering chamber may easily be several orders of magnitude greater than the true reactive scattering yield arriving at the detector from molecules that come directly from the intersection of the reactant beams.

For typical thermal beam intensities, which correspond to reactant concentrations of $\sim 10^{-5}$ Torr in the crossed beam zone, and a reaction cross section of $\sim 10^{-16}$ cm^2, the product intensity corresponds to $\sim 10^{-10}$ Torr in the crossed beam zone and $\sim 10^{-15}$ Torr at the detector (number density about 300 molecules cm^{-3}, flux about 2×10^9 molecules$\cdot$sr$^{-1}\cdot$sec^{-1}). At these concentrations the discreteness of matter is very evident; statistical variations in the signal and background are the limiting source of noise. During a time interval t, the total number of events counted at the detector is $N_1=(S+B)t$ in the open channel and $N_2=Bt$ in the closed channel, with S and B the signal and background, respectively (ion counts per unit time). Since the standard deviation is $N^{\frac{1}{2}}$ in both channels, the counting time required to achieve a given signal-to-noise ratio $R=St/(N_1+N_2)^{\frac{1}{2}}$ is given by $t=R^2(S+2B)/S^2$. This relation emphasizes the need to make both S and S/B as large as possible, in order to make t small enough ($\gtrsim 10^2-10^3$ sec) to allow data to be taken for a wide range of scattering angle and velocity. If $S\ll B$, the averaging time required increases as S^{-2}, whereas if $S\gtrsim B$ it increases only as S^{-1}. In the usual case, $S\ll B$, an increase in signal size, $S\rightarrow S'$ (without increase of B) will decrease t to $t(S/S')^2$, whereas an increase in detector efficiency or a decrease in B (without reducing S) will decrease t in the same proportion.

To maximize the signal size for a given reaction requires intense parent beams and high detection efficiency, to minimize the background requires high mass resolution, use of beam modulation, and drastic pumping of the beam sources and particularly the ionization region. A quadrupole mass filter[16] has an important advantage over magnetic mass analysis in that the transmission and resolution of the filter can be readily adjusted to optimize the conflicting requirements of high detector sensitivity and resolution. Since much lower extraction voltages must be used with a filter, it is more difficult to achieve high efficiency in collecting ions from the ionizer and this disadvantage must be offset by a lens system. The most significant function of beam modulation is that it allows the ambient (unmodulated) background of product species within the detector to be subtracted out unambiguously. As it provides continuous comparison of the open and closed detector channels, the uncertainty in the difference reading is not augmented by the time normalization required to account for long term drifts in the parent beam intensities.

The pumping provided for the ionization region of the detector should (1) reduce the partial pressure of background gas (at the mass of interest) entering from the scattering chamber to a level comparable to the product pressure, i.e., to $\gtrsim 10^{-14}$ Torr and (2) avoid creating background within the ionization region. To achieve (1) in a compact space requires use of two or more stages of differential pumping, with speed sufficient to reduce the flow of background gas to the minimum set by direct line flow

[9] S. Datz and E. H. Taylor, J. Chem. Phys. 39, 1896 (1963).
[10] W. L. Fite and R. T. Brackmann, in *Atomic Collision Processes*, M. R. C. McDowell, Ed. (North-Holland Publishing Co., Amsterdam, 1963), p. 955; J. Chem. Phys. 42, 4057 (1965).
[11] S. Datz and T. W. Schmidt, *Proceedings of the Fifth International Conference on Physical Electronic and Atomic Collisions* (Publishing House Nauka, Leningrad, 1967), p. 247.
[12] Y. T. Lee, J. D. McDonald, P. R. LeBreton, and D. R. Herschbach, J. Chem. Phys. 49, 2247 (1968); 51, 455 (1969).
[13] J. D. McDonald, P. R. LeBreton, Y. T. Lee, and D. R. Herschbach, J. Chem. Phys. (to be published).
[14] D. Beck, F. Engelke, and H. J. Loesch, Ber. Bunsenges. Physik. Chem. 72, 1105 (1968).
[15] J. B. Cross and N. C. Blais, J. Chem. Phys. 50, 4108 (1969).
[16] W. Paul and M. Raether, Z. Physik 140, 262 (1955); W. Paul, H. P. Reinhard, and V. von Zahn, ibid. 152, 143 (1958).

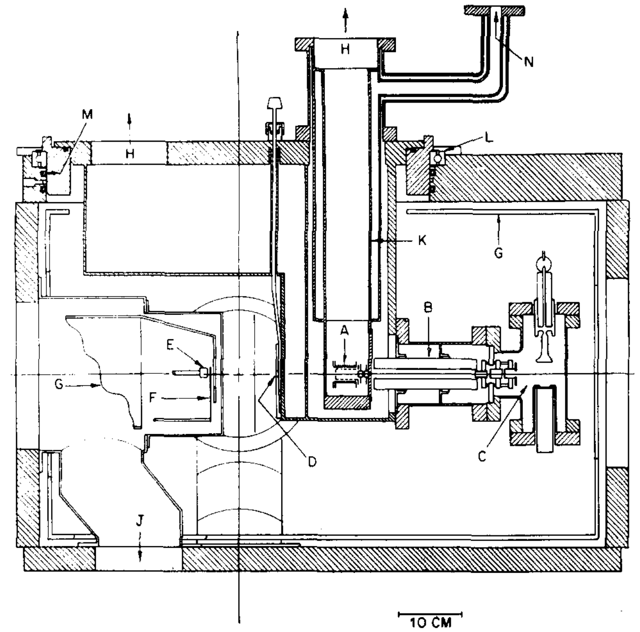

FIG. 1. Side view of apparatus. (A)—electron bombardment ionizer; (B)—quadrupole mass filter; (C)—ion counter; (D)—isolation valve; (E)—molecular beam source; (F)—beam flag; (G)—cold shields; (H)—to ion pumps; (J)—to oil diffusion pump; (K)—liquid nitrogen trap; (L)—ball bearing support of rotatable lid; (M)—rotatable vacuum seal; and (N)—to liquid nitrogen reservoir.

from the crossed beam zone. To ensure (2) requires use of ion, sublimation, or cryogenic pumps and elimination of materials other than metals or ceramics. Also, buildup of layers of product molecules on surfaces near the ionizer must be prevented. This calls for a design such that molecules which pass through the ionizer without being ionized fly on into another differentially pumped region before hitting a surface.

The background problem also places much more stringent requirements on the purity and collimation of the reactant beams than in alkali work. The presence of minute amounts of the product molecule in a parent beam can totally obscure the reactive scattering near the beam direction, due to elastic scattering of the impurity by the other beam which generates a modulated background. This makes it advantageous to modulate the parent beam which is least subject to contamination. Product impurity in the parent beam that is modulated is particularly troublesome, since then elastic scattering by the ambient gas also generates a large modulated background. Strong pumping of the scattering chamber is therefore essential, despite the use of mass analysis and modulation. If the concentration of parent beam species within the scattering chamber is not kept sufficiently low, these species effuse back into the high density region of the beam sources and react there, contaminating the parent beams with the product species. The standard configuration for alkali experiments, in which a well defined weak beam is crossed by a wide intense beam, is often not appropriate with electron bombardment detection. A wide beam aggravates the contamination problems and may raise the background over a broad angular region. In addition, because the viewing angle of the detector is necessarily quite narrow (here 2°), a well defined crossed beam zone is desirable, of width comparable to the viewing angle. Thus, often it is appropriate to use equivalent geometry for the two parent beams even at the cost of decreasing appreciably the joint intensity in the scattering zone.

APPARATUS DESCRIPTION

Figures 1 and 2 show the apparatus plan and typical beam geometry. The scattering chamber is a large box with a rotatable lid which carries the detector unit and its pumping system. The beam sources are placed in separately pumped side chambers which insert into ports in the main chamber. These source chambers and all the beam defining elements (slits, flags, and modulator) are readily interchangeable. The configuration of the apparatus described here is that used in the studies of halogen atom–molecule exchange reactions.[12] For the experiments with deuterium atoms[13] an additional differential pumping chamber (not shown) was installed in the atom beam source.

Vacuum System

The main chamber, a welded stainless steel box with inside dimensions 96×96×76 cm, is pumped by an unbaffled 4200 liter/sec oil diffusion pump (NRS HS-10, with DC-705 silicone fluid) which is mounted to the chamber bottom with a gate valve. Since the alignment of beam sources and detector depends on precisely machined circular rails, the walls are thick (6.4 cm top, 3.7 cm sides and bottom) to prevent distortion by atmospheric pressure. An entire side of the box (facing side in Fig. 1) is removable.

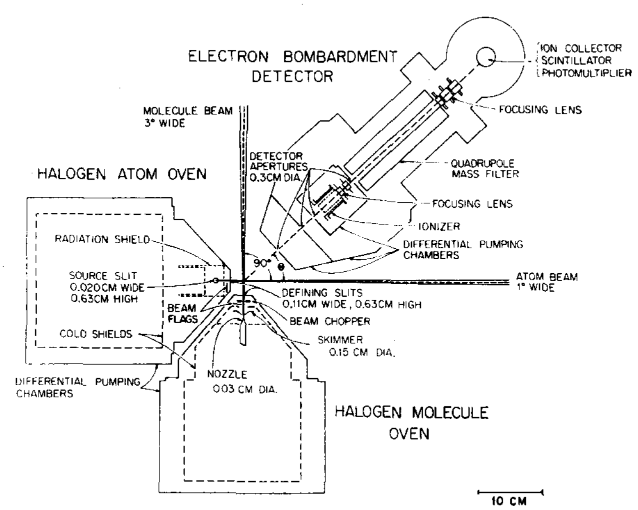

FIG. 2. Top view of apparatus, showing arrangement of differential pumping for detector and source chambers and beam geometry. The scanning range of the detector covers 140°, from 20° beyond the atom beam to 30° beyond the molecule beam.

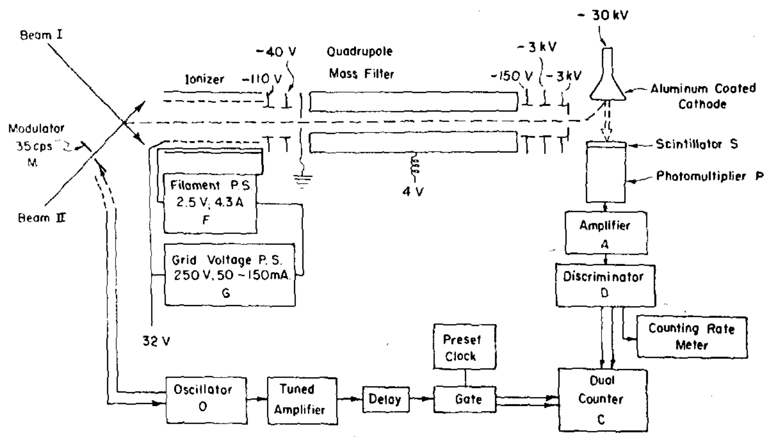

FIG. 3. Block diagram of detector. (A)—amplifier, EG&G model AN 101/N; (C)—dual counter, Transistor Specialty, Inc., 100 Mc model 1535; (D)—discriminator, EG&G model T101/N; (F)—filament power supply, Hewlett-Packard (HP) model 6263; (G)—grid voltage power supply, HP model 890A; (M)—modulator, American Time Product (ATP) type 40 light chopper; (O)—oscillator, ATP model 2HAR-5A; (P)—photomultiplier, Electro-Megadyne, Inc., type 9524S; (S)—plastic scintillator, Pilot Chemical Co., type Pilot B, with 500 Å thick aluminum coating.

The box is lined with a liquid nitrogen cooled shield of 0.6 cm thick copper sheet (nickel plated) which is attached to an external reservoir (Linde CR-25, 25 liters). All the vacuum seals to this chamber are made with greased O-rings.

The rotatable lid is supported by a 63.5 cm i.d. ball bearing (Kaydon Manufacturing Company). The angle scan is made manually with a sprocket-and-chain arrangement (not shown). The vacuum seal is made with two graphite embedded Teflon "Tec Rings" (Tec-Seal Corp.), with the volume between them evacuated by a mechanical pump.

The detector unit has three nested chambers, each pumped by a 50 liter/sec ion pump (Ulteck Corporation, "D-I"). The first region (left-most pump exit H in Fig. 1) serves as a buffer chamber only. The second contains the mass filter and ion detector (B and C; the pump exit is not visible since it is behind the third chamber). The third, innermost chamber is wrapped in a liquid nitrogen trap (K, filled by a 10 liter reservoir at N) and pumped by the third ion pump (right-most pump exit H). The electron bombardment ionizer (A) is mounted within a copper block which is brazed to the trap. This design enables the trap to serve for both the second and third chambers and also allows molecules which go through the ionizer without being ionized to exit from the inner chamber without hitting a surface. Because of the long passageway in contact with liquid nitrogen, it is necessary to allow for thermal expansion (>0.1 cm!) when aligning the ionizer with the mass filter.

The detector unit can be closed off from the main scattering chamber by a slide valve (D), which is sealed with a Viton O-ring. The electrical connections to the detector (not shown in Fig. 1) are made with ceramic-to-metal seals. The ionizer leads go through the second ion pumped chamber; the mass filter and ion counter leads go through the main chamber.

The beam source chambers are each differentially pumped by oil diffusion pumps (NRC VHS-4, with DC-704 silicone fluid) which are mounted to the bottom of the main chamber by gate valves. The source chambers are separated from the main chamber by removable bulkheads made from stainless steel sheet metal. The beam sources are mounted on removable flanges for easy accessibility. Liquid nitrogen traps for cold shields within the sources are also mounted on these flanges and fed by 10-liter reservoirs.

Detector

A block diagram of the detection scheme is shown in Fig. 3, and details of the electron bombardment ionizer in Fig. 4. The ionizer utilizes electron space charge to focus and accelerate the positive ions produced.[4–7] It is based on Brink's design,[6] modified by floating the entire ionizer above ground and by the addition of two focusing lenses (F and G in Fig. 4), which allow the energy and radial velocity of the ions entering the mass filter to be optimized. This improved the transmission (for a given mass resolution) by a factor of 10. The emerging ions are sent through the grounded plate beyond these lenses with just enough

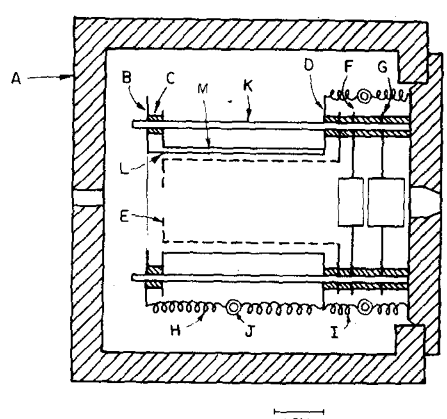

FIG. 4. Electron bombardment ionizer. (A)—chopper chamber, attached to liquid nitrogen trap (Fig. 1); (B)—filament holder; (C)—alumina spacer; (D)—support plate for outer shield; (E)—grid, platinum clad molybdenum; (F)—ion extraction electrode; (G)—focusing lens; (H)—spring to maintain filament tension; (I)—spring holding outer shield; (J)—ceramic ring insulator; (K)—alumina support rod; (L)—thoriated tungsten filament (0.025 cm diam); (M)—outer shield.

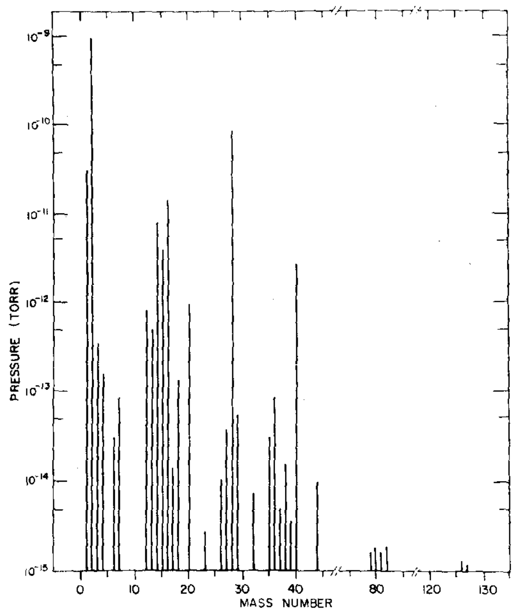

FIG. 5. Mass spectrum of detector background.

energy ($\sim$10–20 eV) to prevent noticeable space charge effects. The quadrupole mass filter is similar to that described by Wilson.[17] The filter has four mass ranges: 1–14, 3–52, 10–140, 30–500. The maximum acceptable ion energy varies from 50 V for the lowest mass range to 5 V for the highest range. The lenses at the exit of the mass filter focus the transmitted ions, which emerge with a substantial spread in radial velocity and position, into a fast beam (3 kV) suitable for acceptance by the ion counter. Calibrations employing a Faraday cup show that this lens system extracts $>95\%$ of the ions.

The ion counting system is based on the design of Daly.[18] The ions are accelerated onto an electrode (aluminum coated stainless steel) held at -30 kV. The angle of incidence (45°) is as far from normal as possible in order to maximize secondary electron emission. The secondary electrons (6–8 per ion) are accelerated in the same 30 kV field and strike an aluminum coated plastic scintillator (S in Fig. 3), producing light pulses (3–6 photons per electron) which are detected by the photomultiplier (P). The discriminator (D) is set to reject the single photon pulses produced by stray light from the filament and thermoelectric emission, while allowing the multiple photon pulses from the scintillator to pass on to the scalers. The over-all efficiency of this ion counter is $\sim$90–95%, almost independent of the ion mass.[18,19] The residual background with no ions incident on the aluminum plate is about 10 counts/sec. Since the photomultiplier is sealed off, the detector is unaffected by exposure to reactive gases. This property, as well as higher gain and stability, makes the scintillation detector preferable to an electron multiplier. It is also preferable, for our application, to a semiconducting[19] detector because the silicon crystal of the latter is susceptible to contamination and because the scintillation detector has a much higher potential counting rate, about 10^7 sec^{-1} (limited by the photomultiplier pulse width, 12 nsec). The only critical aspect of the design is the thickness of the aluminum film on the scintillator. This was deposited by vacuum evaporation, during which the thickness was monitored by observing the transmission of light from the evaporation heater. The film must be thin enough to allow electrons to pass through with small energy loss but thick enough to stop light from the filament from reaching the photomultiplier and to prevent sublimation of the plastic scintillator. A compromise[20] is reached at about 500 Å.

The gating circuitry[21] consists of a clock, operating on the line frequency, which allows total counting times of from 1 to 8000 sec; a comparator to convert the reference signal from the modulator to standard digital logic levels; a monostable multivibrator to provide phase adjustment by delay of the reference signal; a "flipflop" to switch between the chopper-on and chopper-off scalers; and a multivibrator to generate the gating pulses. Use of the same delay to gate both scalers ensures that the on-times are equal within one part in 10^5. The circuit actuates the on-scaler first and the off-scaler last. All the gating circuit components are made of DEC R- and W-series "flip-chip" modules.[22]

Beam Sources

The halogen atoms are formed by thermal dissociation in a graphite tube (0.63 cm diam, wall thickness 0.1 cm). The construction and dimensions of the oven are similar to a previously described hydrogen atom source,[23] except that the mounting block is made of nickel and the current carrying copper tubing is nickel plated to avoid corrosion. The oven is operated at $\sim$1600–1700 K (heating current of 160 A at 2 V) and at halogen gas pressures of 1–2 Torr; the dissociation is $\gtrsim 75\%$ for Cl_2 and $\gtrsim 95\%$ for Br_2. For work with deuterium atoms,[13] the graphite tube is replaced by a tungsten tube operated between 2700 and 2900 K and the differential pumping is augmented with a second stage pumped by a 15 cm oil diffusion pump (NRC VHS 6) mounted on the outer flange of the source chamber. The halogen molecule beam comes from a nozzle source[24] (Fig. 2), operated at 200–500 K and 10–100 Torr. The

[17] K. R. Wilson, UCRL Rept. 11605, Lawrence Radiation Laboratory, University of California, Berkeley, 1964. Similar mass filters are now available commercially, e.g., from Extra-Nuclear Laboratories, Pittsburgh, Pa.
[18] N. R. Daly, Rev. Sci. Instrum. **31**, 264 (1960). See also, H. M. Gibbs and E. D. Commins, ibid. **37**, 1385 (1966).
[19] F. S. Goulding, Nucl. Instrum. Methods **43**, 1 (1966).
[20] J. R. Young, J. Appl. Phys. **27**, 1 (1956).
[21] R. W. Anderson, Ph.D. thesis, Harvard University, 1968.
[22] Digital Equipment Corp., Maynard, Mass.
[23] M. A. D. Fluendy, R. M. Martin, E. E. Muschlitz, Jr., and D. R. Herschbach, J. Chem. Phys. **46**, 2172 (1967).
[24] J. B. Anderson, R. P. Andres, and J. B. Fenn, Advan. Chem. Phys. **10**, 275 (1966).

skimmer for this source and the internal gas feed tubes for both sources are heated (tantalum wire, fiber glass insulation) to overcome radiation loss to the cold shields. In the external glass vacuum line which supplies the source gases, greaseless stopcocks (Teflon–Viton glass[25]) are used.

Originally[12] the modulator was a tuning fork (as shown in Fig. 3), but eventually this proved unreliable due to corrosion and it has now been replaced by a rotating drum chopper. The drum (6 cm high and 0.6 cm diam) rotates about its vertical axis and the beam passes through a 0.4 cm wide diametral slot. A hysteresis motor[26] drives the drum. The chopping frequency can be varied from 20 to 200 Hz; it is usually 141 Hz, but for heavy molecules such as I_2 it is reduced to as low as 35 Hz to avoid time-of-flight effects.

PERFORMANCE

The effectiveness of the pumping system is illustrated in Fig. 5, which shows the mass spectrum of the detector background in the absence of the parent beams, i.e., with the slide valve (D in Fig. 1) closed, and in Fig. 6, which indicates the total pressures and the partial pressures of reactants and products (BrCl or DBr) in various regions. The detector background contains, in addition to peaks at mass numbers less than 44 which are typically found in high vacuum systems, peaks arising from residual amounts of hydrogen halides, HCl, HBr, and HI. Elsewhere the background spectrum falls below 10^{-15}–10^{-16} Torr. In the experiments done to date, which all involve a halogen molecule or a hydrogen halide or deuterium halide as the product,[27] the increase in the unmodulated background at the product mass which is observed on opening the slide valve is usually of the same order of magnitude or less than the background with the slide closed.

In Fig. 6, the total background refers to uncorrected ionization gauge readings. The partial pressures in the source chamber were measured directly by an ionization gauge and in detector region 3 by the mass spectrometer (intensity scale normalized to the ion pump gauge total pressure reading). Partial pressures in the main chamber were estimated indirectly by measuring the increment in intensity as the slide valve between the main chamber and detector was opened and closed and dividing this increment by the geometrical viewing factor of the ionizer (0.012 sr). Partial pressures in detector regions 1 and 2 were interpolated using the known geometrical factors and pumping speed.

The valving procedure was also used to calibrate the over-all sensitivity of the detector. This measurement was made with a known pressure of nitrogen at 300 K (cold traps empty) introduced into the main chamber. When the mass filter transmission is unity, about one in 1000 molecules incident on the detector is counted. This corresponds to 10^{18} counts·sec^{-1}·Torr^{-1} (0.2 A·Torr^{-1}) at the ionization region. For complete resolution of adjacent masses ($<10^{-5}$ crosstalk) the transmission of the mass filter is unity up to mass 20 but it falls off to 0.1 at mass 100. Except for reaction systems which require discrimination between adjacent masses, however, in experiments the resolution of the mass filter is set to allow collection of all isotopes of the product species and the transmission is then essentially unity.

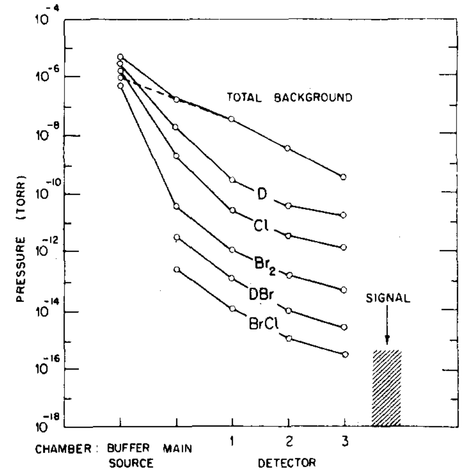

FIG. 6. Comparison of intensity of various background components in different regions with reactive scattering signal for the $Cl+Br_2$ reaction (detected product BrCl) and the $D+Br_2$ reaction (detected product $D^{81}Br$). Total background is indicated with (——) and without (- - -) the parent beams "on"; other curves refer to the "on" case.

Angular distributions of reactive scattering have been measured[12,13] with this apparatus (in operation for 18 months) for 16 systems from five families: $X+YZ$, $D+XY$, $X+HY$, $D+HX$, and $Cl+D_2$ (where X, Y, Z denote halogen atoms, Cl, Br, or I). Velocity analysis of the product has also been carried out for six of these reactions by means of a time-of-flight analyzer, as described elsewhere.[13] Rate constants determined from conventional kinetic studies indicate activation energies ranging from zero to about 5 kcal/mole and total reaction cross sections of about 10^{-17} to 10^{-15} cm^2 for these systems.

Table I lists counting times, background with slide valve closed (B_0) and open (B), and signal-to-noise (S/N) ratios for a few reactions which illustrate the range of operating conditions. These data refer to the peak of the product distribution. In most cases, it was possible to obtain satisfactory data, typically $R \lesssim 5$ at least for $t < 10^3$ sec, down to signal levels of a few counts per second. The relation

[25] West-Glass Corp., El Monte, Calif.

[26] Type CC, Globe Industries (Dayton, Ohio). The power supply comprises a Hewlett-Packard (Palo Alto, California) type 200 CD oscillator and a Dyna Co. (Philadelphia, Pa.) Dynaco mark III amplifier.

[27] For reactions producing DBr or DI the deuterium labeling effectively circumvents the residual hydrogen halide background, but for $D^{37}Cl$ there is a troublesome background, due primarily to the higher vapor pressure, to the tail of the very large Ar^+ peak at mass 40, to K^+ from the ionizer filament, and to Br^{2+}.

TABLE I. Detector operating conditions for various reactions.

Reactants	Detected product	Exptl.[a]	t sec	B_0 counts/sec	B counts/sec	S	R
$Cl+Br_2$	BrCl	AD	4	100	500	600	28
		VA	40	8	50	80	35
$D+Br_2$	$D^{81}Br$	AD	20	200	1100	130	12
		VA	200	15	40	10	15
$D+I_2$	DI	AD	200	30	90	10	8
$D+HI$	DI	AD	10	40	1400	600	39
$D+Cl_2$	$D^{37}Cl$	AD	100	1700	2000	70	11
$Cl+HI$	$H^{37}Cl$	AD	40	13 000	14 000	200	7
$Cl+DI$	$D^{37}Cl$	AD	20	1500	2000	200	12

[a] AD denotes an angular distribution measurement, VA velocity analysis.

expected for statistics limited noise, $t=R^2(S+2B)/S^2$, was checked for several systems and found to hold accurately. Since the time-of-flight velocity analysis decreases B as well as S, it does not impair the S/N ratio if t is increased in proportion to the closed-to-open ratio of the chopper, a factor of 10. Furthermore, for product velocities that are appreciably higher than the mean thermal velocity (at liquid nitrogen temperature), the analysis discriminates against the background; e.g., the velocity analysis reduces B/S by $\sim35\%$ for $Cl+Br_2$ and $\sim100\%$ for $D+Br_2$.

The $Cl+Br_2$ reaction is the most favorable case found thus far, by virtue of a large cross section, absence of any interfering background mass peaks, and the convenient vapor pressure of Br_2 (high at the source temperature and low at liquid nitrogen temperature) which enables use of a very intense beam. For $D+Br_2$ the reaction cross section is probably similar but the higher mass resolution required reduces the transmission to $\sim20\%$. For reactions of Cl, Br, or D with I_2 the cross sections are comparable to reactions with Br_2 but a 10-fold less intense beam had to be used. Reactions which produce DCl or HCl were the most difficult to study because of the high residual background.[27] The worst case studied involves HCl from $Cl+HI$; however, despite $B/S\sim70$–2200 over the accessible range of the product distribution, the signal level (with mass-filter transmission $\sim40\%$) was high enough to provide satisfactory data without unduly long counting times.

The presence of a modest activation energy is an advantage in these experiments. This can markedly reduce the background generated from the ambient reactant gases, which are at liquid nitrogen temperature, whereas it does not much reduce the product yield from the scattering zone, since the mean collision energy of the parent beams is ~3 kcal/mole for halogen atoms (at 1700 K) and ~5 kcal/mole for H or D atoms (at 2700 K). In the study of $D+X_2$ reactions,[13] there appeared near ($\pm10°$) the parent X_2 beam a large and obviously spurious signal, attributed to reaction with background D atoms modulated by the beam chopper. This spurious peak was very large for the I_2 and Br_2 reactions, for which $E_{act}<1$ kcal/mole, but absent altogether for Cl_2, which has $E_{act}\sim3$ kcal/mole.

The quality of the data obtained with this apparatus for reactions of $\sim10^{-15}$ cm^2 cross section is as good as that obtained for alkali reactions with a surface ionization detector. For reactions with cross sections of $\sim10^{-16}$ cm^2 or less, it is considerably better, since the surface ionization method depends on differencing the total and nonreactive scattering signals. Except at angles fairly near the parent beam (in most cases within about $\pm60°$), however, accurate measurements of nonreactive scattering with the electron bombardment detector are often more difficult than measurements of the reactive scattering, since (as illustrated in Fig. 6) the background of beam materials in the main chamber is much larger than the background of product. Also, measurements very near the modulated parent beam (within $\gtrsim5°$) encounter difficulty with buildup of space charge within the ionizer, which produces a substantial (5%) modulation of all background mass peaks. The most serious limitation of this apparatus, for both reactive and nonreactive scattering, is the fact that the residual detector background remains high ($\lesssim10^{-13}$ Torr in Fig. 5) for about half of the mass numbers below 40, almost prohibitively high ($\lesssim10^{-11}$ Torr) for about 10% of those mass numbers. This can probably be most effectively reduced by use of a liquid helium cryosorption pump.

ACKNOWLEDGMENTS

We are much indebted to Dr. R. W. Anderson, who designed the gating circuit, and to G. Pisiello, whose skillful and zealous aid greatly accelerated the construction of this apparatus. We are very grateful for financial support received from the National Science Foundation and from the William F. Milton Fund of Harvard University.